



Thermal elastohydrodynamic simulation of involute spur gears incorporating mixed friction

Lars Bobach*, Ronny Beilicke, Dirk Bartel, Ludger Deters

Otto von Guericke University Magdeburg, Institute of Machine Design, Chair for Machine Elements and Tribology, Universitaetsplatz 2, 39106 Magdeburg, Germany

ARTICLE INFO

Article history:

Received 25 August 2011

Received in revised form

25 November 2011

Accepted 29 November 2011

Available online 7 December 2011

Keywords:

Spur gears

Transient thermal elastohydrodynamics

Surface roughness

Three-dimensional contact

ABSTRACT

A model for calculating transient, three-dimensional, thermal elastohydrodynamic tooth flank contacts in spur gears with involute gearing is presented. The calculation model is based on the combined numerical solutions of the generalized Reynolds, energy and Fourier heat equations. Mass conserving cavitation, non-Newtonian flow and the real involute characteristics of the tooth flanks are incorporated. States of mixed friction and microhydrodynamic effects are ascertained integrally based on real-measured surface topographies. Straight spur gear pairs serve as an example to present the results of calculating the influence of surface roughness asperities and gearing geometry.

© 2011 Elsevier Ltd. All rights reserved.

1. Introduction

Spur gears are implemented in a transmission to transmit and gear torques and rotary motions between parallel shafts (see Fig. 1). The involute has been established to be a common tooth profile. The rolling contact between the tooth flanks has a curvature that varies with the direction of the addendum. A combined sliding and rolling motion always exists on the tooth flanks, with the exception of the pitch point. The amounts of sliding motion reach their maximums in the regions of the addendum and root. The lubricated tooth flank contact constitutes a thermal elastohydrodynamic line contact that expands finitely in the direction of the width of the tooth flanks. Because velocity, curvature and load change as the contact point moves along the tooth flanks, the operating conditions are transient, regardless of the external operating parameters.

The tribological processes at the tooth surface contact greatly influence the operational reliability and efficiency of a spur gear. Significant variables of these processes are the pressures and shear stresses occurring in the contact and the resultant temperatures. High pressures and shear stresses can cause fatigue in the tooth flanks; high temperatures can cause warm scuffing. Calculating the thermal elastohydrodynamic tooth surface contact requires simultaneously analyzing the hydrodynamics in the lubricating gap, the elastic deformations of the contact partners and the thermal conditions in the lubricating gap and solid

bodies. In addition, the lubricant's rheological properties and the affect of mixed friction and microhydrodynamics must be included in these equations. Furthermore, a sound calculation requires an exact description of the real kinematic and geometric relationships of the tooth surface contact and load sharing among the pairs of teeth with multiple meshing.

The development of models that calculate lubricated rolling contact has been a priority of tribology research for many years. The Reynolds equation is applied to calculate hydrodynamic pressure distribution and the theory of the elastic half-space can be applied to incorporate elastic deformation. Normally, the energy equation is applied to ascertain the thermal conditions of the lubricant, and the Fourier heat equation is used to determine the same conditions for the solid bodies. The entire problem and the individual sub-problems can only be completely solved numerically. Dowson and Higginson presented initial solutions for the isothermal elastohydrodynamic contact [1]. The contact was simplified to a line contact that expanded infinitely in the direction of width. This finding reduces the problem to a planar, two-dimensional contact. Because the contact of the two tooth flanks is normally longer in the direction of width than in the direction of the addendum, this simplification has been adopted for most models that calculate spur gears.

In 1981, Wang and Cheng presented an early, comprehensive study on a numerical simulation of the contact conditions of straight spur gear pairs [2,3]. This group calculated the transient, elastohydrodynamic tooth surface contact along the length of action, which was simplified two-dimensionally on the basis of the numerical approximation functions from Vichard [4]. Models have been introduced that determine the dynamic load sharing of

* Corresponding author. Tel.: +49 391 67 12648; fax: +49 391 67 11166.
E-mail address: lars.bobach@ovgu.de (L. Bobach).

Nomenclature

b	width (m)
c	specific heat capacity (J/kgK)
E_{red}	reduced Young's modulus (N/m ²)
f	coefficient of friction (dimensionless)
F_n	normal force (N)
g_α	length of action (m)
h	film thickness (m)
h_0	nominal film thickness (m)
h_{cr}	critical film thickness (m)
h_{def}	deformed gap height (m)
h_{min}	minimum film thickness (m)
$h_{\delta w}$	local deformed gap height (m)
k	coefficient of thermal distribution (dimensionless)
O	center point (dimensionless)
p	pressure (N/m ²)
p_{cav}	cavitation pressure (N/m ²)
p_e	base pitch (m)
$p_{\text{c,lim}}$	plastic flow pressure (N/m ²)
$\dot{q}$	heat flux density (W/m ²)
r	radius (m)
s_{y0}	coordinate of the length of action (m)
t	time (s)
u, v, w	velocities in x, y and z direction (m/s)
u_d	relative velocity (m/s)
w	deformation (m)
x, y, z	Cartesian coordinates (m)
Y_0	contact point (dimensionless)
α	pressure angle (Grad)
α_w	working pressure angle (Grad)

α_y	profile angle (Grad)
β_{th}	coefficient of volume expansion (1/K)
$\dot{\gamma}$	shear rate (1/s)
δ_y	tooth trace variation in the direction of width (m)
η	dynamic viscosity (Pas)
η^*	effective dynamic viscosity (Pas)
θ	gap fill factor (dimensionless)
ϑ	temperature (K)
λ	thermal conductivity (W/mK)
ρ	density (kg/m ³)
τ	shear stress (N/m ²)
τ_0	boundary shear stress, Eyring (N/m ²)
τ_f	friction shear stress (N/m ²)
Φ^p	pressure flow factor (dimensionless)
Φ^s	shear flow factor (m)
ψ_y	roll angle (Grad)
ω	angular velocity (1/s)
Ω	calculation domain (m ²)

Frequently used indices

c	solid body contact
el	elastic
gas/vap	gas–vapor phase
h	hydrodynamics
inv	involute
liq	fluid phase
mix	mixed variable for cavitation
pl	plastic
solid	solid bodies

multiple meshing and calculate the bulk and flash temperatures. Simple finite element models, already used to an extent, were applied. As is common in elastohydrodynamic calculations, the parabolas of each contact point were used to approximate the contact of the tooth flanks. Lin and Medley presented a model to calculate straight spur gear pairs, which was also based on the approximation functions of Vichard [5]. Unlike Wang and Cheng, this group regarded the contact as isothermal and disregarded

dynamic load sharing. However, their calculation model includes a method that ascertains the change of the tooth flanks' radii of curvature, which is strongly variable in the root region. Oster presented a calculation model that was comparable to the aforementioned models [6]. However, the tooth surface contact, which is variable as a function of the length of action, was simplified to a sequence of independent and stationary, two-dimensional elastohydrodynamic contacts. The

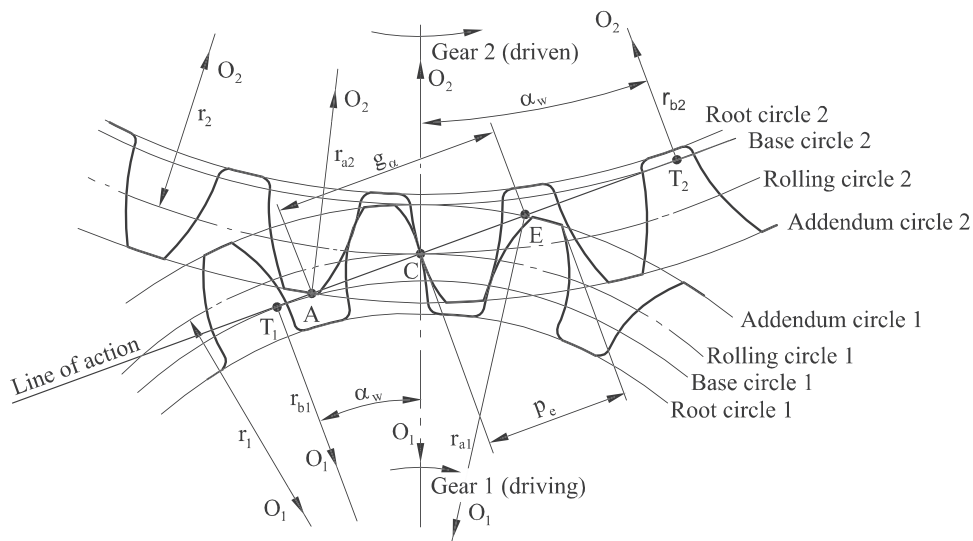


Fig. 1. Geometric conditions of involute gearing.

model includes empirical approaches to calculating friction and temperature, and the stress properties in the materials beneath the contact surface were calculated.

Larsson presented the first complete, numerical solution for transient, two-dimensional elastohydrodynamic tooth surface contact in 1997 [7]. The calculation model from this study is confined to an isothermal approach but incorporates non-Newtonian flow as a shear thinning fluid model with a shear stress maximum. As was mentioned for the previous models, in Larsson's model, second order parabolas were utilized to approximate tooth flank profiles. Because pairs of teeth alternate between single and multiple meshing, the loading along the length of action is specified as cascaded. These results elucidate the transient effects, that occur when alternating single and multiple meshing. Wang et al. specified a numerical solution for transient thermal elastohydrodynamic tooth flank contact [8]. Furthermore, they simplified the contact two-dimensionally and assumed that the flow was Newtonian. The results from this study demonstrate that markedly lower minimum film thicknesses but only slightly higher maximum pressures can be expected in thermal calculations. Li and Kahraman calculated the two-dimensional elastohydrodynamic tooth surface contact and incorporated real measured profiles of the roughness of contact surfaces [9]. This group implemented an algorithm that allowed for direct asperity contacts. However, their calculations were confined by the assumption of isothermal conditions. The results presented in [10] also incorporate surface roughness; however, the calculation model is based only on approximations that specify the elastohydrodynamics of and statistical models for the contact problem. Castro and Seabra likewise presented a model for the calculation of tooth flank contacts under mixed friction conditions [11]. This model incorporates for the planar, two-dimensional contact real-measured profiles of the roughness of the surfaces. With the aid of an empirical load sharing function, the pressure distributions along the length of action of the gearing are calculated individually from the elastohydrodynamics and pure solid body contacts and are subsequently superimposed. The shear stresses of the boundary friction are approximated with the aid of a constant boundary friction coefficient.

The aforementioned studies treated the tooth surface contact in the direction of width solely as a planar, two-dimensional contact and assumed infinite expansion. However, the calculation model must be extended to three dimensions to ascertain the influence of the commonly employed tooth trace correction in the direction of width, e.g. crowning, filleting or chamfering, as well as misalignment between the gear wheels on the tribological processes in the contact. A model that calculates the elastohydrodynamics of three-dimensional elliptical point contacts to analyze the effects of mixed friction in tooth flank contacts has been introduced by Holmes et al. [12]. The model is applied solely to select quasi-static operating points of the meshing. The meshing along the entire length of action cannot be calculated transiently; however, the calculation model allows for the employing of real-measured surface topographies and captures local, transient effects that results from the movement of the rough surfaces. Steady-state, three-dimensional calculations that are based on a calculation model have been introduced by Ren et al. [13] are also presented by Zhu et al. [14], but these only calculate the point where commencing single meshing is critical. However, no models that calculate a complete, numerical solution for transient, three-dimensional thermal elastohydrodynamic tooth surface contacts have been established.

Apart from the spur-gear problem, various numerical solutions have been developed for geometrically simple, three-dimensional point and line contacts. Hamrock and Dowson [15], Yang and Wen [16], Venner [17], Wijnant [18], Chevalier

et al. [19], Solov'yev [20] and Scholz [21] have all published studies on point contacts. Yang and Wen introduced an expanded Reynolds equation that facilitated non-Newtonian and thermal calculations with a variable viscosity and density of the fluid in the direction of gap height. Venner developed a multigrid method that expedited the numerical solution and established the basis for calculating highly resolved elastohydrodynamic contacts. Wijnant and Chevalier implemented a model that was developed by Elrod [22] that incorporates mass conserving cavitation. Solov'yev presented a thermal elastohydrodynamic solution for the rough point contact and Scholz factored in the influence of transient effects on smooth and rough surfaces. Three-dimensional calculations for elastohydrodynamic line contacts were presented isothermally by Mostofi and Gohar [23] and Park and Kim [24] and thermally by Liu and Yang [25] and Sun and Xiaoyang [26]. These calculations were performed for the stationary contact of a cylindrical roller against a plane with a fluid with Newtonian flow properties. A transient, three-dimensional model of elastohydrodynamic line contacts was introduced in [27]. Taking the contact inner race/spherical roller of a spherical roller bearing as an example, results were presented, which allowed for non-Newtonian flow, states of mixed friction, microhydrodynamic effects and lubricant starvation.

The following paper presents a numerical calculation model that simulates, three-dimensionally, the thermal elastohydrodynamic tooth surface contact in straight spur gears. The tooth surface contact is calculated transiently along the entire length of action. Load changes that result from the alternation between single and multiple meshing and the variable, tangential velocities of the tooth flanks are incorporated. The gap profile is derived directly from the real involute profiles of the tooth flanks at every time step. In addition, an allowance can be made for tooth trace corrections, e.g. crowning, filleting or chamfering, in the direction of the width of the tooth flanks. The calculation model is based on the solution of the generalized Reynolds equation and includes algorithms that ascertain mass-conserving cavitation and describe the non-Newtonian flow of the fluid. The temperature distribution is calculated by simultaneously solving the transient energy equation for the lubricating gap and the transient Fourier heat equation for the solid bodies. An allowance is made for microcontacts in the area of mixed friction by integrating integral solid body contact pressure curves and microhydrodynamic effects are integrated in the calculation model as flow factors. Both models are based on real measured surface topographies of the tooth flanks. Straight spur gear pairs serve as an example to present the results of the calculation.

2. Calculation model

The calculation model that is presented in the following sections is part of the calculation software Tribo-X, which is a proprietary development by the authors and simulates various tribological contacts.

2.1. Geometry, load and kinematics of meshing

The tooth flank profiles of involute spur gears are designed as involutes of a circle and are engineered by unrolling a straight line onto the base circle of the particular spur wheel. By definition, an intermeshed pair of teeth shares a fixed normal at every contact point Y_0 , which is tangential to both base circles (see Fig. 2). The contact point moves with constant velocity on this normal, which is referred to as the line of action. Meshing begins at point A, intersects

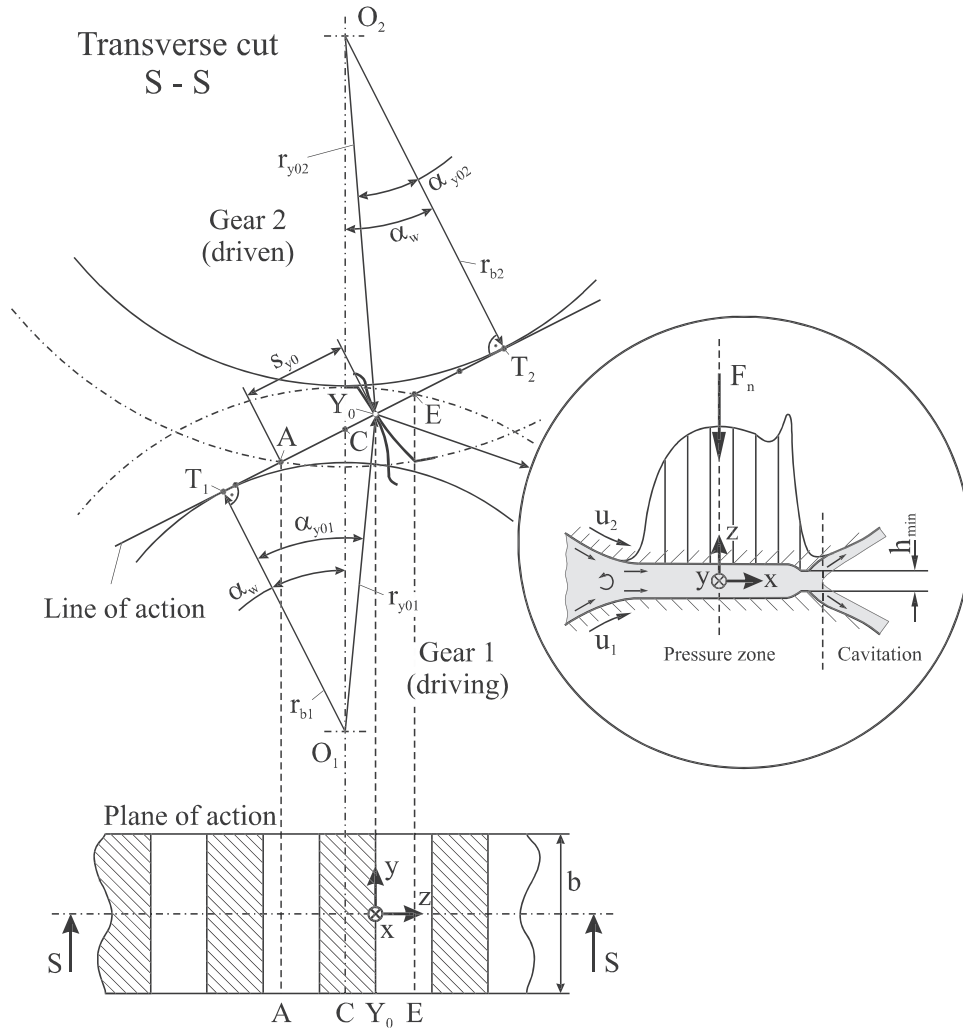


Fig. 2. Meshing of a straight spur gear pair with involute gearing.

the driven gear wheel's addendum circle with the line of action, and ends at point E, which is the intersection of the addendum circle of the driving gear with the line of action. Starting from point A, the coordinate s_{y0} is introduced along the length of action and defines the geometric location of the contact point (see Fig. 2).

The tooth surface contact is calculated along the entire length of action from point A to point E. During its motion along the length of action, the contact point is characterized by variable loads, velocities and tooth flank profiles. Thus, a transient problem exists. The variable load is produced by the alternating meshing of several pairs of teeth. Straight spur gear pairs, when practically designed, normally alternate between single and double meshing. However, uniform load sharing between the pairs of double meshing teeth cannot be assumed because of the elastic deformations of the intermeshed teeth. Hence, the characteristic with a load, which is linear variable in double meshing and is represented in Fig. 3, is usually assumed for gearing without any profile correction.

The tangential velocities of the tooth flanks produce the effective, hydrodynamic velocity of the elastohydrodynamic tooth surface contact. These velocities are variable along the line of action and, with the constant angular velocities of the driving and driven gear wheels ω_1 and ω_2 , respectively, are defined using the following equations:

$$u_1 = \omega_1(T_1\bar{A} + s_{y0}) \quad u_2 = -\omega_2(AT_2 - s_{y0}) \quad (1)$$

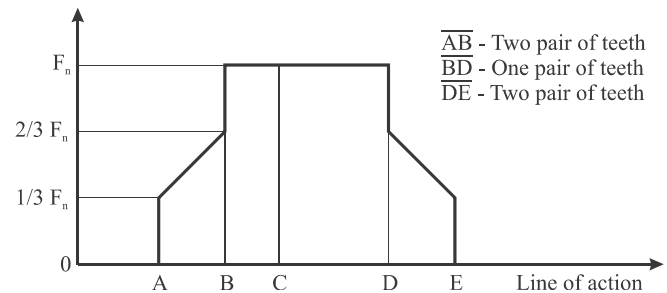


Fig. 3. Load sharing without profile correction (schematic).

The value of the resultant sliding, relative velocity between the tooth flanks $|u_d| = |u_1 - u_2|$ equals zero at the pitch point C, increases on both sides as the distance from the pitch point grows linearly and, depending on the location of the pitch point, has its maximum at point A or E.

The gap geometry of the active contact area is described with the aid of a local Cartesian coordinate system (x, y, z) that moves with the contact (see Fig. 2). The coordinate system's origin is located at the contact point Y_0 . The vertical z -axis is perpendicular to the tooth flanks. The curves of the involutes define the tooth flank profiles at the contact point. An involute curve changes its curvature constantly as the distance to the base circle increases. In the calculation model, the real involute curves map the tooth

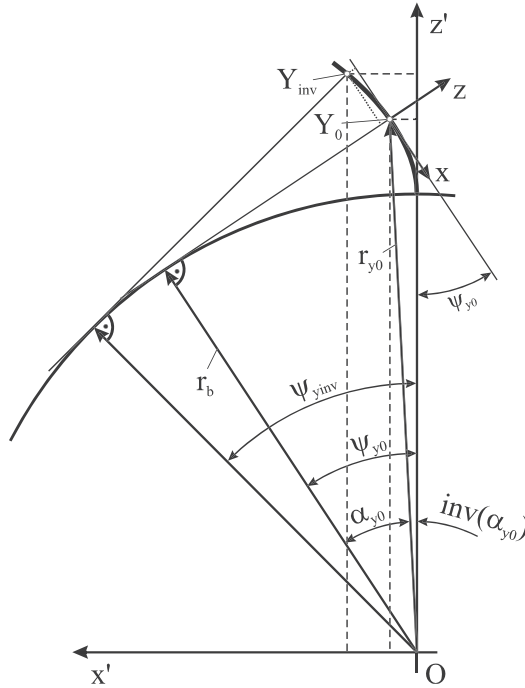


Fig. 4. Coordinate transformation of the involute function.

flank profiles for any position of the contact point on the length of action. Therefore, the involute curves must be described in the x - z plane of the local Cartesian coordinate system. Because it moves along the involute during meshing, the curve of the involute is initially described in a fixed plane coordinate system (x' , z') with its origin at the center of the gear wheel (see Fig. 4). The location of any point Y_{inv} on the involute is defined in the fixed coordinate system (x' , z') as a function of the roll angle ψ_{yinv} :

$$x'_{inv}(\psi_{yinv}) = r_b(\sin(\psi_{yinv}) - \widehat{\psi}_{yinv} \cos(\psi_{yinv})) \quad (2)$$

$$z'_{inv}(\psi_{yinv}) = r_b(\cos(\psi_{yinv}) + \widehat{\psi}_{yinv} \sin(\psi_{yinv})) \quad (3)$$

Transferring the point Y_{inv} to the x - z plane of the local Cartesian coordinate system necessitates first calculating the location of the contact point Y_0 , relative to the center of the gear wheel as the profile angle α_{y0} and the radius r_{y0} :

$$\alpha_{y0} = \arctan\left(\frac{T_1 A + S_{y0}}{r_b}\right) \quad r_{y0} = \frac{r_b}{\cos(\alpha_{y0})} \quad (4)$$

Using the involute angle $\text{inv}(\alpha_{y0}) = \tan(\alpha_{y0}) - \widehat{\alpha}_{y0}$, the position of Y_0 in the fixed coordinate system (x' , z') is calculated as the following equation:

$$x'_0 = r_{y0} \sin(\text{inv}(\alpha_{y0})) \quad z'_0 = r_{y0} \cos(\text{inv}(\alpha_{y0})) \quad (5)$$

Eqs. (2), (3) and (5) and the roll angle $\widehat{\psi}_{y0} = \tan(\alpha_{y0})$ can be used to transform point Y_{inv} into the x - z plane of the local coordinate system (x , y , z):

$$x_{inv}(\psi_{yinv}) = -x'_{inv} \sin(\psi_{y0}) - z'_{inv} \cos(\psi_{y0}) + x'_0 \sin(\psi_{y0}) + z'_0 \cos(\psi_{y0}) \quad (6)$$

$$z_{inv}(\psi_{yinv}) = -x'_{inv} \cos(\psi_{y0}) + z'_{inv} \sin(\psi_{y0}) + x'_0 \cos(\psi_{y0}) - z'_0 \sin(\psi_{y0}) \quad (7)$$

In the calculation model, the coordinate z_{inv} is sought for a given coordinate x_{inv} . The roll angle ψ_{yinv} is calculated iteratively from the Eqs. (2), (3), (5) and (6) and inserted into Eq. (7). The aforementioned approach can be used to calculate the involute

profiles of both gear wheels for any contact point on the length of action in the local coordinate system. Readers are referred to the relevant literature for the rules of calculating the underlying variables of geometric gearing that are not mentioned here [28].

2.2. Elastohydrodynamics

Fig. 2 presents a schematic of the pressure distribution and the lubricating gap properties of an elastohydrodynamic tooth surface contact in the direction of movement (x direction). When incorporating the elastic deformations of the solid bodies, the involute profiles of both gear wheels z_{inv1} and z_{inv2} following Eq. (7), the defined tooth trace variations in the direction of the tooth width δ_{y1} and δ_{y2} (see Figs. 8 and 17) and a constant of integration h_0 , the film thickness characteristic in the x and y directions for any point on the length of action can be calculated with the following equation:

$$h(x, y) = h_0 + z_{inv1}(x) + z_{inv2}(x) + \delta_{y1}(x, y) + \delta_{y2}(x, y) + \frac{1}{\pi E_{red}} \iint_{(\Omega)} \frac{p_h(x', y') dx' dy'}{\sqrt{(x-x')^2 + (y-y')^2}} \quad (8)$$

The calculation of the elastic deformations is based on the theory of the elastic half-space [29]. The basis of the elastohydrodynamic calculation model is established by the following generalized Reynolds equation for compressible and laminar flow with an isotropic, non-Newtonian fluid at a temporally variable film thickness and variable density and dynamic viscosity in the x , y and z directions [29]:

$$\frac{\partial}{\partial x} \left(\Phi_{xx}^p G_{1x} \frac{\partial p_h}{\partial x} \right) + \frac{\partial}{\partial y} \left(\Phi_{yy}^p G_{1y} \frac{\partial p_h}{\partial y} \right) = \frac{\partial}{\partial x} \left(\theta G_{2x} \frac{(u_2 - u_1)}{F_0} + \theta G_3 u_1 + \theta G_3 \Phi_{xx}^s \frac{(u_2 - u_1)}{2h} \right) + \frac{\partial}{\partial t} (\theta G_3) \quad (9)$$

with $p_h > p_{cav}$ and $\theta = 1$ in the pressure zone

$p_h = p_{cav}$ and $\theta < 1$ in the cavitation region

$$G_{1x} = G_{1y} = \int_0^h \rho_{liq}(z) \left[\int_0^z \frac{z'}{\eta_{liq}^*(z')} dz' - \frac{F_1}{F_0} \int_0^z \frac{1}{\eta_{liq}^*(z')} dz' \right] dz$$

$$G_{2x} = \int_0^h \rho_{liq}(z) \left[\int_0^z \frac{1}{\eta_{liq}^*(z')} dz' \right] dz \quad G_3 = \int_0^h \rho_{liq}(z) dz$$

$$F_0 = \int_0^h \frac{1}{\eta_{liq}^*(z)} dz \quad F_1 = \int_0^h \frac{z}{\eta_{liq}^*(z)} dz$$

Non-Newtonian flow is expressed as a nonlinear correlation between the hydrodynamic shear stress τ_{fh} and the shear rate $\dot{\gamma}$ in the lubricating gap. This action is described by replacing the fluid's dynamic viscosity η_{liq} with a variable, effective viscosity η_{liq}^* in the z direction, which locally retains the formulation of Newton's law of shear stress [16,29]:

$$\dot{\gamma}_{zx}(z) = \frac{\partial u}{\partial z} = \frac{\tau_{fhx}(z)}{\eta_{liq}^*(z)} \quad \text{and} \quad \dot{\gamma}_{zy}(z) = \frac{\partial v}{\partial z} = \frac{\tau_{fhy}(z)}{\eta_{liq}^*(z)} \quad (10)$$

Eyring's shear thinning fluid model [30] is applied to represent the non-Newtonian flow:

$$\dot{\gamma} = \frac{\tau_0}{\eta_{liq}} \sinh\left(\frac{\tau_e}{\tau_0}\right) \quad \text{with} \quad \tau_e = \sqrt{\tau_{fhx}^2 + \tau_{fhy}^2} \quad (11)$$

Following Eq. (11), the shear rates in the x and y directions of the contact that are introduced in Eq. (10) can be formulated:

$$\dot{\gamma}_{zx}(z) = \frac{\partial u}{\partial z} = \frac{\tau_{fhx}(z)}{\eta_{liq}(z)} \frac{\tau_0}{\tau_e(z)} \sinh\left(\frac{\tau_e(z)}{\tau_0}\right) \quad (12)$$

$$\dot{\gamma}_{zy}(z) = \frac{\partial v}{\partial z} = \frac{\tau_{fhy}(z)}{\eta_{liq}(z)} \frac{\tau_0}{\tau_e(z)} \sinh\left(\frac{\tau_e(z)}{\tau_0}\right) \quad (13)$$

Assuming that the hydrodynamic shear stresses in the z direction of the lubricating gap develops linearly

$$\tau_{fzx}(z) = \tau_{fzx,1} + z \frac{\partial p_h}{\partial x} \quad \text{and} \quad \tau_{fzy}(z) = \tau_{fzy,1} + z \frac{\partial p_h}{\partial y} \quad (14)$$

Eqs. (12) and (13) are integrated as a function of the film thickness h . The solution of the resultant equation system, which is based on the unknown shear stresses at the surface 1 $\tau_{fzx,1}$ and $\tau_{fzy,1}$, is used to calculate the curves of the shear stresses according to Eq. (14) and to determine the curve of the effective dynamic viscosity from the Eqs. (10) and (11):

$$\eta_{liq}^*(z) = \eta_{liq}(z) \frac{\tau_e(z)}{\tau_0} \sinh\left(\frac{\tau_e(z)}{\tau_0}\right)^{-1} \quad (15)$$

Readers are referred to [31] for more details on the implementation of this model.

The correlations of density with pressure [32] and temperature [29] were determined. A model presented by Rodermund [33] was utilized to ascertain the correlation between the viscosity and pressure and temperature. When the masses are balanced, the validity of the Reynolds equation presented in Eq. (9) extends to the entire calculation domain of the lubricating gap and, thus, includes regions of pressure and cavitation. Cavitation arises at the elastohydrodynamic contact in the region of the gap outlet where the quantity of fluid that is flowing out no longer suffices to completely fill the expanding lubricating gap (see Fig. 2). The cavitation region is characterized by a rupturing of the lubricant film and the development of a two-phase flow, which consists of a fluid and a gas–vapor phase. An algorithm that is based on the JFO theory to describe the flow characteristics in the cavitation region is applied to the calculation model [22]. The Reynolds equation is expanded by a gap fill factor θ , which expresses the ratio of the lubricant volume in the cavitation region as $0 < \theta < 1$. The insertion of the gap fill factor in the form presented transforms the flow in the cavitation region into a homogeneous, two-phase flow. Thus, the Reynolds equation is formulated to be a complementary problem for the unknown variables of the hydrodynamic pressure p_h and the gap fill factor θ . Furthermore, the Reynolds Eq. (9) contains pressure and shear flow factors (explained in Section 2.3), which capture microhydrodynamic effects. The hydrodynamic pressure distribution must be in equilibrium with the external load F_n , so that force equilibrium is calculated by the following equation:

$$F_n = \int_{(\Omega)} p_h(x,y) d\Omega \quad (16)$$

2.3. Mixed friction and microhydrodynamics

Mixed friction denotes the operating state of simultaneously occurring fluid and boundary friction as well as hydrodynamic pressures and solid body contact pressures at a lubricated contact with rough surfaces. Mixed friction begins when the thickness of a film is fallen below the critical film thickness h_{cr} and, thus, the first microcontacts between the roughness asperities of the contact surfaces develop (see Fig. 5).

Given the current state of computing power, the size of the domain that is needed to calculate spur gears does not allow for resolution that is high enough to directly map the rough surfaces in the film thickness curve with sufficient precision. Hence, a discrete model of elastic/plastic solid body contact, which is based on a previous study [29], is first used to determine local solid body contact pressures of representative sections of the real measured surface topography distributions (see Fig. 6). The model of solid body contact is a separate calculation module. Building upon the theory of the elastic half-space, the model calculates the distributions of contact pressure and deformation. Moreover, any surface topography that is desired can be specified. Simplified elastic/plastic deformation is incorporated with the aid of a linear/elastic ideal/plastic material model. The maximum pressures that occur at the contact are, therefore, confined to a plastic flow pressure $p_{c,lim}$, which is to be defined. In keeping with this approach, elastic and plastic deformations are also distinguished. The amount of the plastic flow pressure is determined using microhardness measurements. Boundary conditions of cyclic deformation are specified at the edges to calculate the representative surface sections. When a defined approximation of the bodies is specified, the distribution of contact pressure and deformation is calculated iteratively by transforming the equations of the elastic half-space into a discrete system of linear equations. Furthermore, the multilevel multi-integration, which was described by Venner [17], is applied to expedite the calculation.

With the aid of the contact model, integral solid body contact pressures p_c are derived from local distributions as a function of the mean deformed gap height h_{def} between the representative rough surfaces (see Fig. 7). The variables p_c and h_{def} are defined as the mean values of the distributions of the discrete pressure and gap height within the representative surface sections, respectively. The incremental approximation of both rough surfaces in the solid body contact model delivers the functional curve of p_c as a function of h_{def} . This curve can be used to determine the integral solid body contact pressure for any film thicknesses in the actual model for the thermo-elastohydrodynamic calculation of tooth flank contact. A pressure distribution then appears in the mixed friction area of the tooth flank contact, which results from the

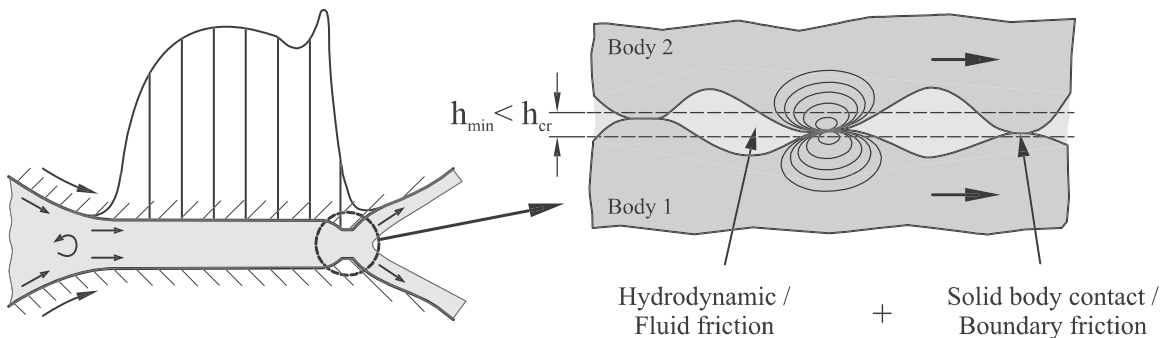


Fig. 5. Mixed friction.

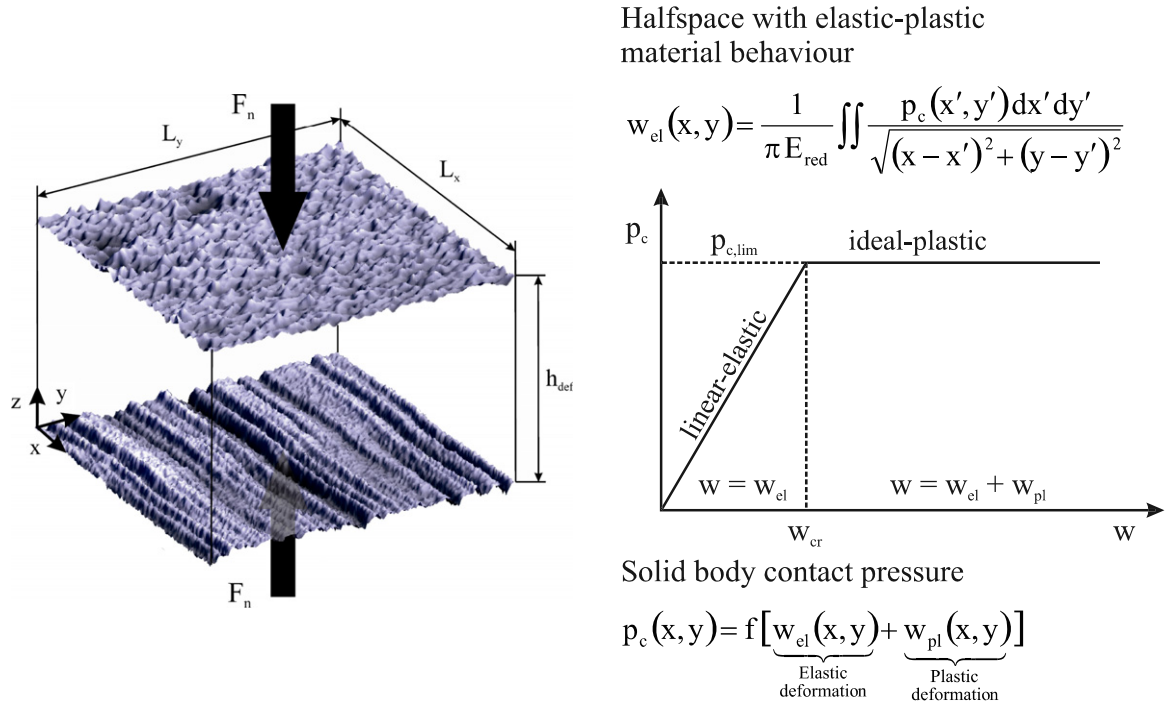


Fig. 6. Solid body contact model.

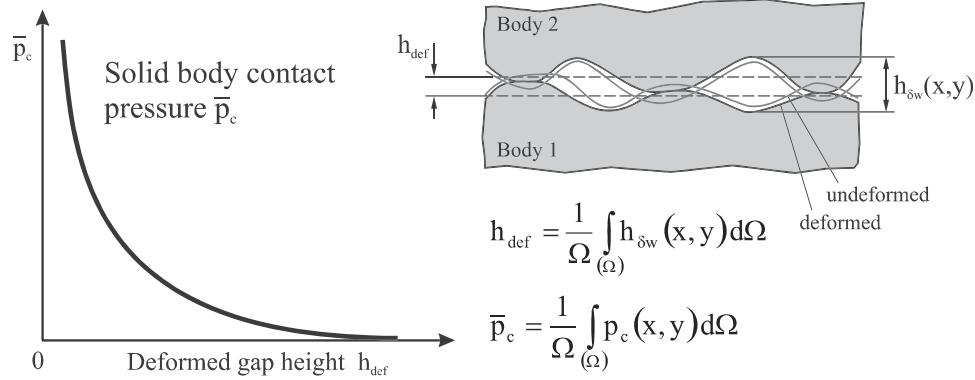


Fig. 7. Integral solid body contact pressure (schematic).

aggregate distributions of the hydrodynamic pressures and the integral solid body contact pressures:

$$p(x, y) = p_h(x, y) + \bar{p}_c(x, y) \text{ with } \bar{p}_c(x, y) = f[h(x, y)] \quad (17)$$

The hydrodynamic shear stress and the shear stress of the boundary friction produce the resultant shear stress distribution in the direction of movement of the contact (in the x direction):

$$\tau_{fx}(x, y) = \tau_{fhx}(x, y) + \bar{\tau}_{fc}(x, y) \text{ with } \bar{\tau}_{fc}(x, y) = f_c \bar{p}_c(x, y) \quad (18)$$

The calculation of the shear stress of the boundary friction is simplified based on the definition of a constant boundary friction coefficient f_c , which is derived from empirical values. The equilibrium of forces, which is based on Eq. (16), expands to the following equation:

$$F_n = \int_{(\Omega)} p(x, y) d\Omega = \int_{(\Omega)} (p_h(x, y) + \bar{p}_c(x, y)) d\Omega \quad (19)$$

The influences of microhydrodynamic effects on the development of hydrodynamic pressure can no longer be disregarded when the film thicknesses are in the range of the size of the roughness asperities of the contact surfaces. Building upon the

studies of Patir and Cheng [34], the Reynolds Eq. (9) was expanded by two pressure flow factors and one shear flow factor to allow for microhydrodynamic effects:

$$\Phi_{xx}^p = \frac{1}{\frac{\rho_{liq} h_{def}^3}{12\eta_{liq}} \frac{\partial \bar{p}_h}{\partial x}} \left(\frac{1}{\Omega} \int_{(\Omega)} \frac{\rho_{liq} h_{dw}^3}{12\eta_{liq}} \frac{\partial p_h}{\partial x} d\Omega \right) \quad (20)$$

$$\Phi_{yy}^p = \frac{1}{\frac{\rho_{liq} h_{def}^3}{12\eta_{liq}} \frac{\partial \bar{p}_h}{\partial y}} \left(\frac{1}{\Omega} \int_{(\Omega)} \frac{\rho_{liq} h_{dw}^3}{12\eta_{liq}} \frac{\partial p_h}{\partial y} d\Omega \right) \quad (21)$$

$$\Phi_{xx}^s = \frac{-2}{(u_2 - u_1) \bar{\theta} \rho_{liq}} \left(\frac{1}{\Omega} \int_{(\Omega)} \frac{\rho_{liq} h_{dw}^3}{12\eta_{liq}} \frac{\partial p_h}{\partial x} d\Omega \right) \quad (22)$$

The flow factors are also calculated numerically on the basis of real measured surface topographies. The microhydrodynamic calculation model that is employed is based on the discrete solid body contact model that is described above. Local solid body contact pressures, the resulting deformations and the boundary conditions of cyclic deformation are incorporated. The flow is

simulated by coupling the contact model with the solution of the Reynolds equation. Based on the pressure and shear flow simulation, specific boundary conditions are specified at the inflow and outflow boundaries (see [34]). Following Eqs. (20)–(22), the flow factors are determined from the calculated microhydrodynamic pressure distributions to be a function of the deformed gap height h_{def} . The existing film thickness is then used to integrate the Reynolds Eq. (9) of the thermo-elastohydrodynamic tooth flank contact. Readers are referred to a previous report [29] for a more detailed description of the calculation model.

2.4. Temperature

The three-dimensional temperature distribution in the lubricating gap is calculated with the aid of the transient energy equation for the fluid, which is used to establish the equilibrium of heat sources and sinks (23). Fluid friction and compression processes act as heat sources. Convection, heat conduction and expansion processes constitute heat sinks. The temperature distribution is essentially dictated by the fluid friction, as a source, and the heat conduction in the gap height direction, as a sink. In addition, the influence of compression by the lubricant as another source of heat increases in transient calculations.

$$\underbrace{\beta_{\text{th,mix}} \vartheta \left(\frac{\partial p_h}{\partial t} + u \frac{\partial p_h}{\partial x} + v \frac{\partial p_h}{\partial y} \right)}_{\text{Compression/Expansion}} + \underbrace{\tau_{\text{fthx}} \frac{\partial u}{\partial z} + \tau_{\text{fthy}} \frac{\partial v}{\partial z}}_{\text{Fluid Friction}} + \underbrace{c_{p,\text{mix}} \rho_{\text{mix}} \left(\frac{\partial \vartheta}{\partial t} + u \frac{\partial \vartheta}{\partial x} + v \frac{\partial \vartheta}{\partial y} + w \frac{\partial \vartheta}{\partial z} \right)}_{\text{Convection}} - \underbrace{\lambda_{\text{mix}} \left(\frac{\partial^2 \vartheta}{\partial x^2} + \frac{\partial^2 \vartheta}{\partial y^2} + \frac{\partial^2 \vartheta}{\partial z^2} \right)}_{\text{Heat Conduction}} \quad (23)$$

Models that were introduced by Larsson and Andersson [35] are applied to ascertain the correlations of the thermophysical variables of the lubricant with the pressure and temperature. In the form that is presented, the energy equation applies to the entire calculation domain of the lubricating gap. The properties of the lubricant in the cavitation region are described with the aid of a mixing approach [29]:

$$\rho_{\text{mix}} = \rho_{\text{gas/vap}} + \theta(\rho_{\text{liq}} - \rho_{\text{gas/vap}}) \approx \theta \rho_{\text{liq}} \quad (24)$$

$$c_{p,\text{mix}} = c_{p,\text{gas/vap}} + \theta(c_{p,\text{liq}} - c_{p,\text{gas/vap}}) \quad (25)$$

$$\lambda_{\text{mix}} = \lambda_{\text{gas/vap}} + \theta(\lambda_{\text{liq}} - \lambda_{\text{gas/vap}}) \quad (26)$$

$$\beta_{\text{th,mix}} = \beta_{\text{th,gas/vap}} + \theta(\beta_{\text{th,liq}} - \beta_{\text{th,gas/vap}}) \quad (27)$$

The mixing approach that underlies the aforementioned equations is based on the assumption that a homogeneous, two-phase flow in the cavitation region exists and, thus, that the variables of the fluid and gas–vapor phases are a function of the gap fill factor θ .

The temperature distribution in the lubricating gap and the shear stresses of the boundary friction in the mixed friction area provide the basis for specifying the temperatures at the surfaces and inside the solid bodies using the transient Fourier heat equation:

$$\underbrace{c_{v,\text{solid}} \rho_{\text{solid}} \left(\frac{\partial \vartheta}{\partial t} \right)}_{\text{Transient Term}} = \underbrace{\lambda_{\text{solid}} \left(\frac{\partial^2 \vartheta}{\partial x^2} + \frac{\partial^2 \vartheta}{\partial y^2} + \frac{\partial^2 \vartheta}{\partial z^2} \right)}_{\text{Heat Conduction}} \quad (28)$$

Solid bodies and lubricant are interrelated as a function of the heat flux densities at the surfaces:

$$\dot{q}_{\text{liq (mix)}} + k|u_d|\tau_{\text{fc}} = \dot{q}_{\text{solid}} \quad \text{with } \dot{q} = \lambda \frac{\partial \vartheta}{\partial z} \quad (29)$$

The transition condition between fluid and solid bodies is supplemented by an additional heat flux density that is the product of the coefficient of thermal distribution k , the relative velocity $|u_d| = |u_1 - u_2|$ and the shear stress of the boundary friction τ_{fc} to ascertain the additional heating of the solid bodies that results from the boundary friction during mixed friction. The coefficient of thermal distribution describes the relative amount of heat flux density in a particular body; thus, $k_1 + k_2 = 1$ ultimately applies to both bodies. Because an exact calculation of the local coefficient of thermal distribution as a function of the length of action would require inordinate computing times, the initially approximated coefficient of thermal distribution is determined only from the thermo-physical characteristics of the solid bodies, which is based on work by Charron:

$$k_1 = \frac{b_{\text{th1}}}{b_{\text{th1}} + b_{\text{th2}}} \quad \text{with } b_{\text{th}} = \sqrt{\lambda_{\text{solid}} c_{v,\text{solid}} \rho_{\text{solid}}} \quad (30)$$

When physical characteristics are identical, a coefficient of thermal distribution of $k_1 = k_2 = 0.5$ is produced for both bodies.

2.5. Numerics

The Reynolds Eq. (9) is solved numerically, incorporating the mass conserving cavitation model following an iterative method proposed by Venner [17]. High numerical efficiency is attained by rigorously applying multigrid methods. The equations that calculate the elastic deformations of the bodies, the rheological properties of the lubricant and the characteristics of the mixed friction and microhydrodynamics are solved simultaneously. In parallel, the energy Eq. (23) and the Fourier heat Eq. (28) are also solved iteratively with the aid of finite difference discretization [20]. Time is integrated incrementally along the length of action, which is subdivided into discrete time steps. The equilibrium of the instantaneous, external load is calculated at every time step [36].

3. Presentation and discussion of the results

A straight spur gear pair with a positive addendum modification (V-plus) and the geometric and operating parameters that are presented in Table 1 was selected for a demonstration calculation. A tooth trace correction, which was in the form of crowning in the direction of width of the tooth flanks, was initially specified (see Fig. 8).

Table 1

Straight spur gear pair V-plus.

Modulus	m	=	4 mm
Pressure angle	α	=	20°
Number of teeth gear 1	z_1	=	19
Number of teeth gear 2	z_2	=	37
Gear ratio	i	=	1.947
Center distance	a	=	115 mm
Addendum modification gear 1	x_1	=	0.6557
Addendum modification gear 2	x_2	=	0.1639
Width of gear 1	b_1	=	25 mm
Width of gear 2	b_2	=	20 mm
Crowning gear 1	δ_{sph1}	=	5 μm
Crowning gear 2	δ_{sph2}	=	5 μm
Clearance factor	c^*	=	0.25
Transverse contact ratio	ε_α	=	1.364
Length of action	g_α	=	16.103 mm
Torque gear 1	T_1	=	100 Nm
Speed gear 1	n_1	=	1485 min ⁻¹
Lubricant	FVA 3		
Lubricant feed temperature	ϑ_{en}	=	80 °C
Bulk temperature of the gears	ϑ_{B}	=	80 °C

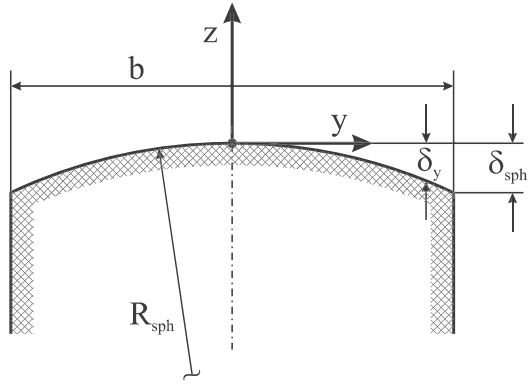


Fig. 8. Crowning.

The given values of the torque and speed produced the characteristics of the load and the tangential and sliding velocities that are presented in Fig. 9 (see Section 2.1). To improve comparability, these and all subsequent characteristics are plotted as a function of a standardized length of action. The parameters for the calculation of the rheological and thermophysical variables of the lubricant as a function of temperature and pressure were obtained from previous studies [29,37,35]. The thermophysical variables of the solid bodies were based on the following values for steel: $\rho = 7840 \text{ kg/m}^3$, $c_p = 460 \text{ J/(kgK)}$ and $\lambda = 46.5 \text{ W/(mK)}$. The boundary shear stress for Eyring's shear thinning fluid model was set at $\tau_0 = 5 \text{ MPa}$.

The tooth flanks of an operated spur gear pair, which were constructed of quenched and tempered steel, were measured three-dimensionally to apply the models that were described in

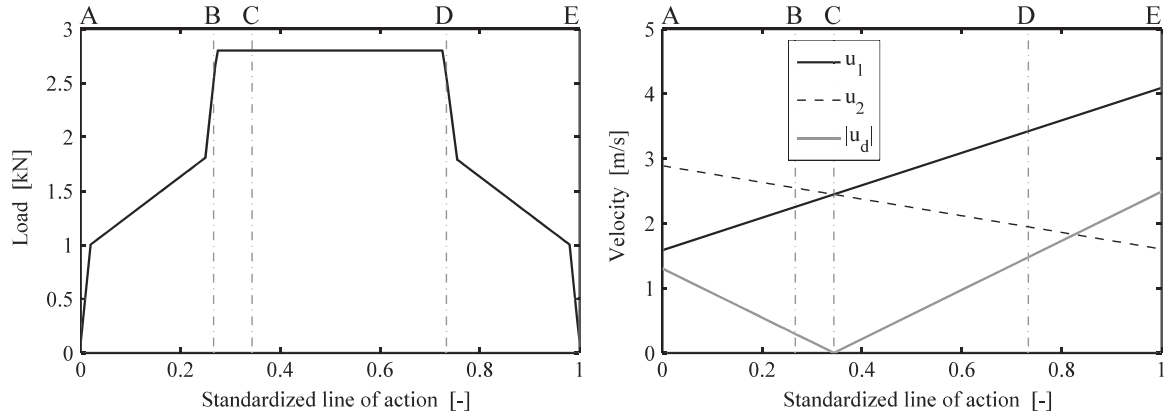
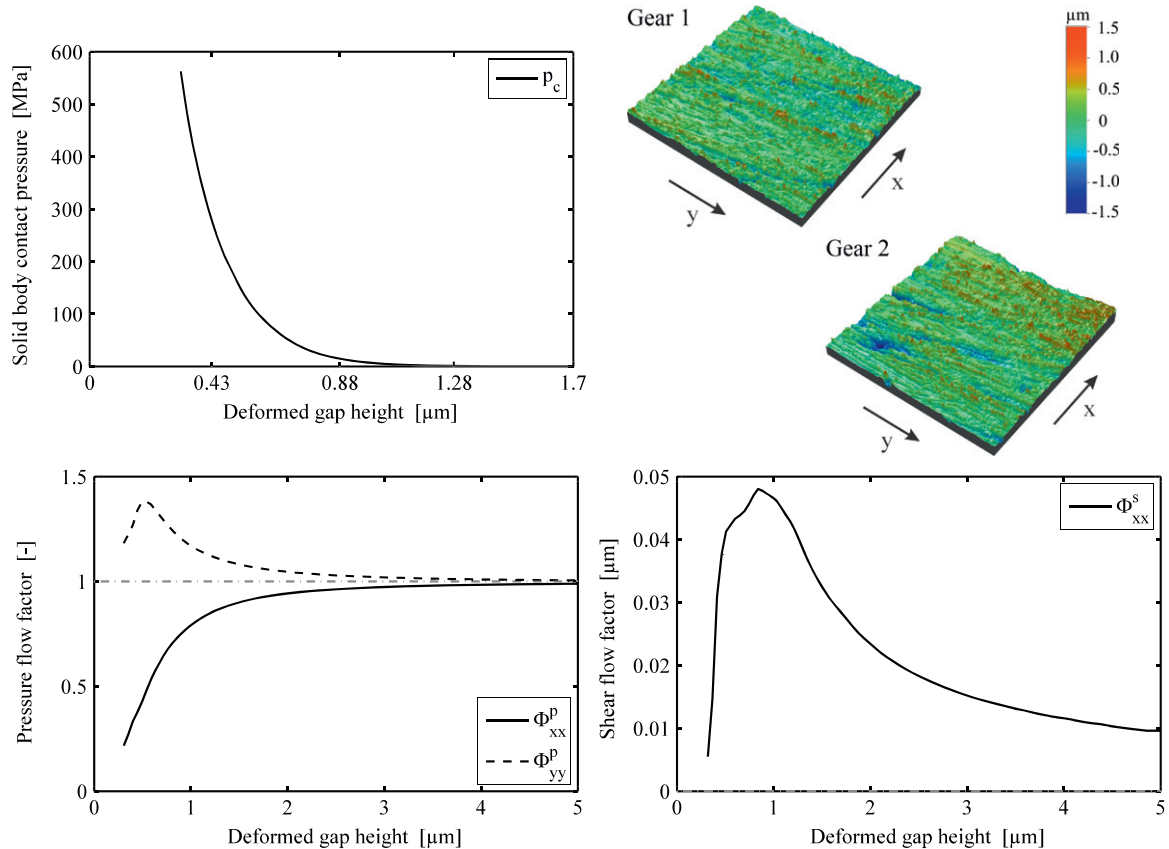
Fig. 9. Load and velocities of the V-plus spur gear pair ($x_{1,2} > 0$).

Fig. 10. Surface topographies, integral solid body contact pressure and flow factors.

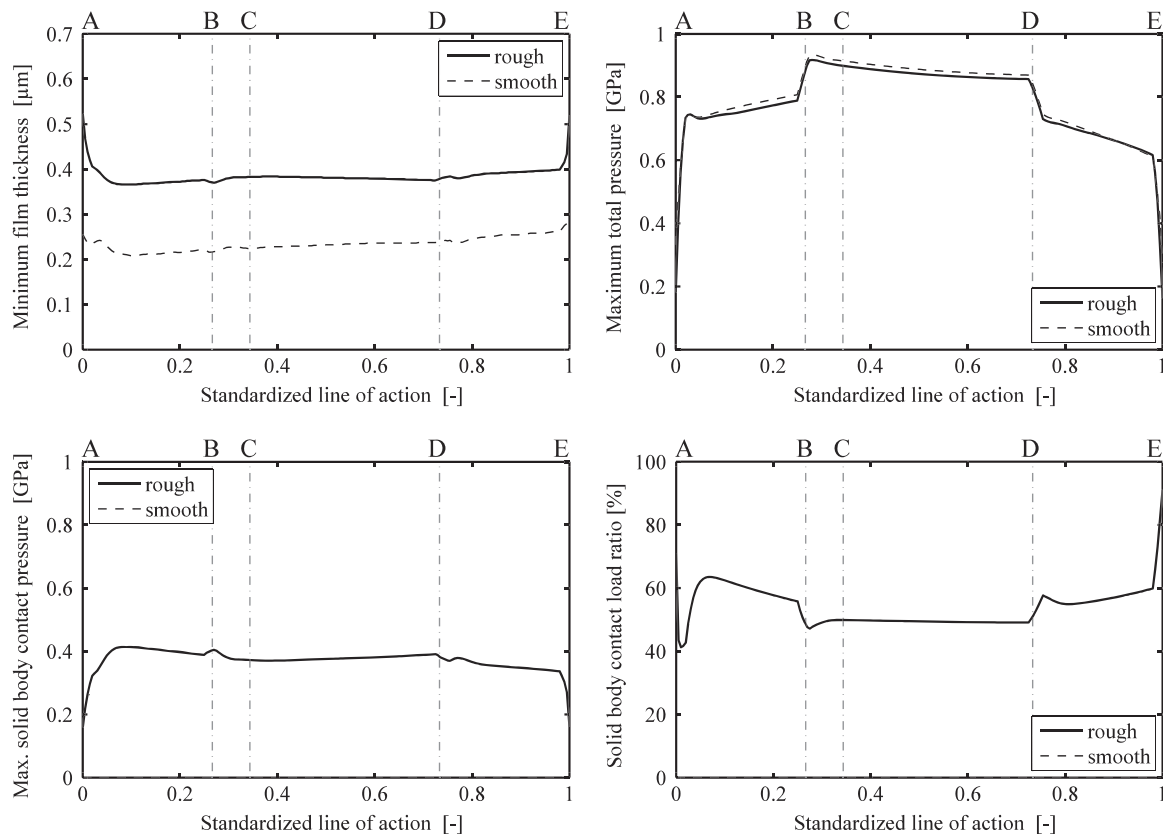


Fig. 11. Results of the V-plus spur gear pair, comparing rough and smooth (part 1).

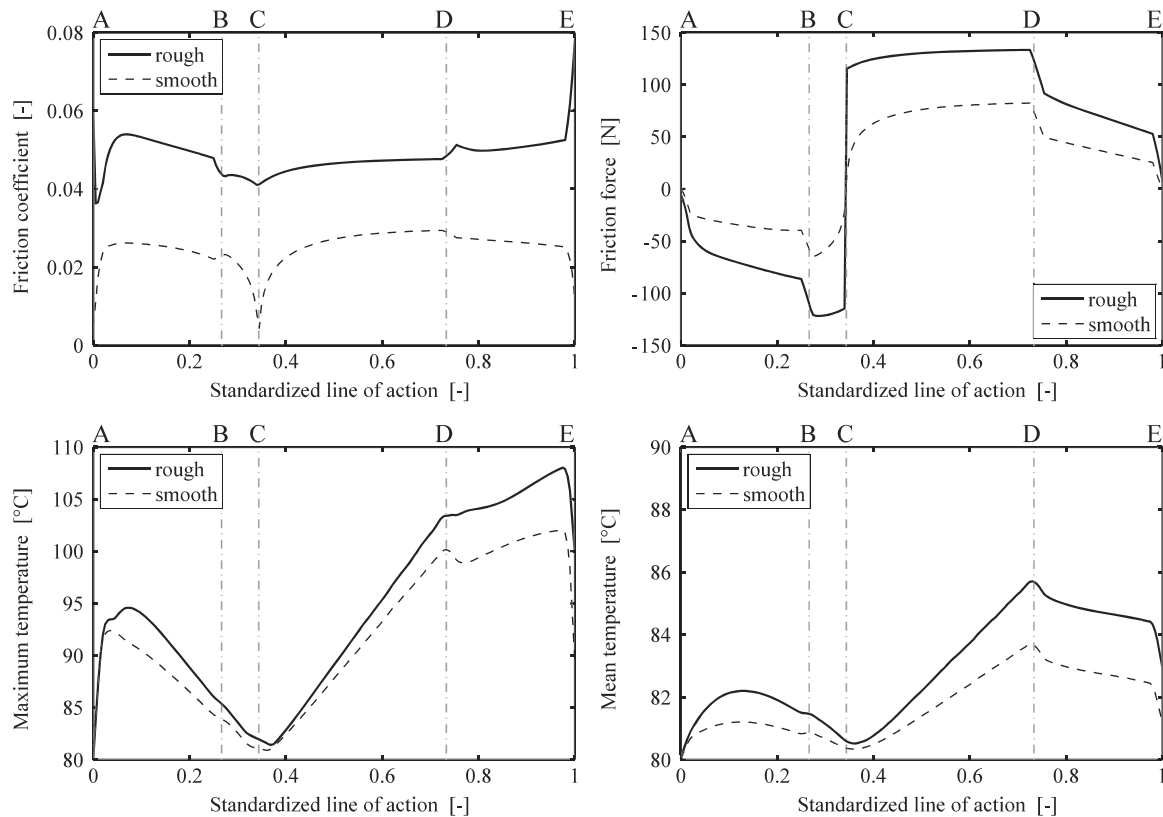


Fig. 12. Results of the V-plus spur gear pair, comparing rough and smooth (part 2).

Section 2.3 to mixed friction and microhydrodynamics (see Fig. 10). The resolution was 249×249 points at intervals of $0.99 \mu\text{m}$. The measurements delivered a critical film thickness of $h_{\text{cr}} = 1.70 \mu\text{m}$ for the transition to mixed friction. The characteristics of the integral solid body contact pressures and flow factors were calculated to be a function of the deformed gap height and are presented in Fig. 10. A plastic flow pressure of $p_{\text{c,lim}} = 4500 \text{ N/mm}^2$, which was ascertained from measurements of Martens hardness, was estimated to be the condition for plastic flow. The integral solid body contact pressures rose nonlinearly as the gap height decreased. The influence of the scoring structure of the surface topographies as an impediment to the flow in the x direction and an improvement of the flow in the y direction was discernible in the curves of the pressure flow factors. In the mixed friction area, the shear flow factor reached its maximum and then decreased again because of the increase of solid body contacts. The coefficient of boundary friction, which determines the shear stress of the boundary friction based on Eq. (18), was set at $f_c = 0.08$.

The calculations of tooth surface contact were performed at a resolution of $129 \times 273 \times 30$ points (x – y – z direction). In the z direction, the number of points is uniformly dispersed on the lubricating gap and both solid bodies. The solid bodies were discretized to a depth of 0.275 mm , with variable point intervals, and the size of the calculation domain was set to 1 mm in the x direction and to 20 mm in the y direction, which corresponds to the width of the driven gear. Time was integrated at 201 discrete time steps along the length of action.

Figs. 11 and 12 present results of the spur gear pair V-plus, which was introduced in Table 1, along the length of action. Calculations that do and do not allow for surface roughness asperities are compared. In both calculations, the minimum film thickness increased slightly after dropping sharply at the start of meshing along the line of action because of the steady improvement of the ratios of curvature in the contact. Aside from brief fluctuations at the transitions, no significant differences are recognizable between the characteristics of single and double

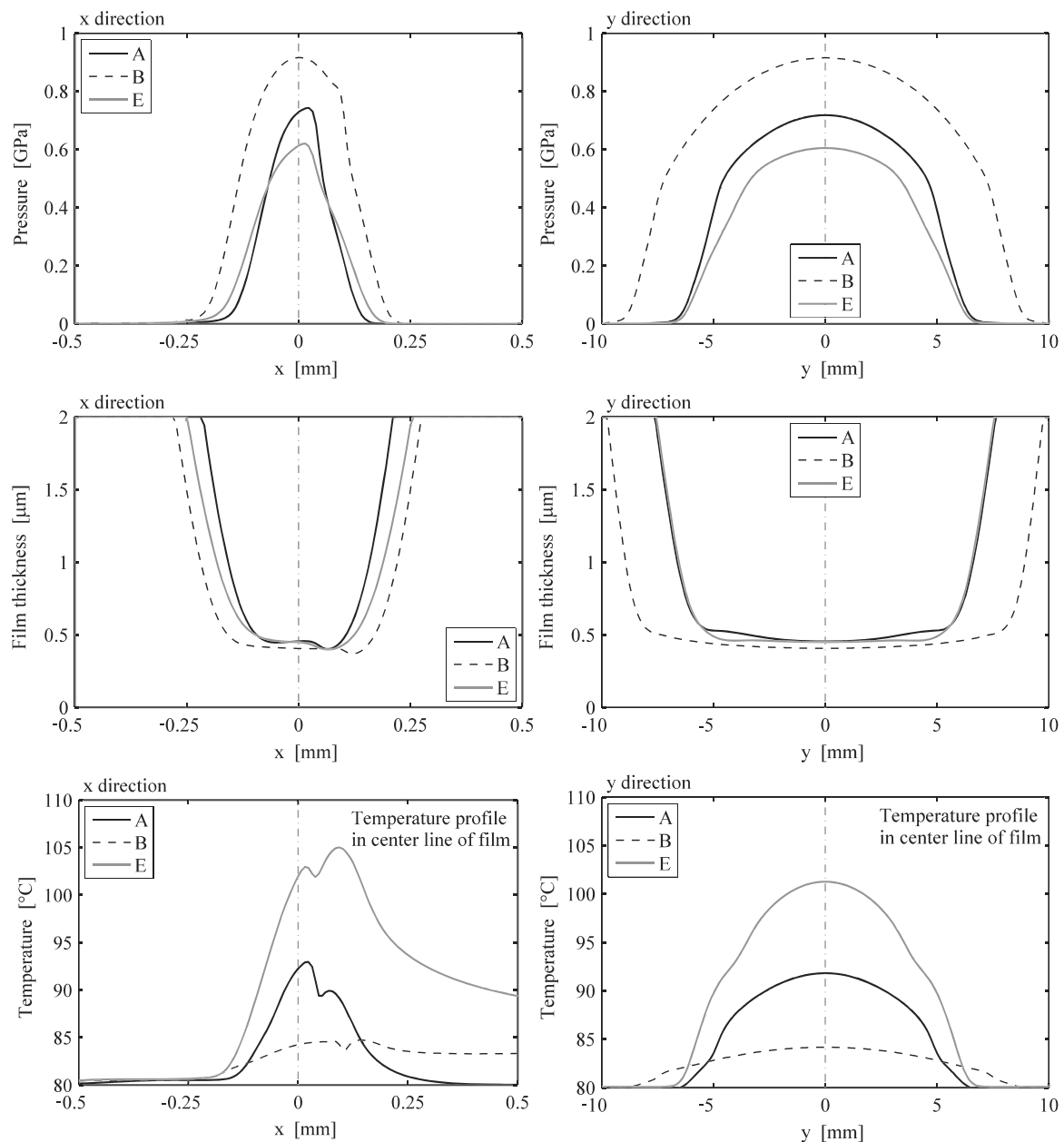


Fig. 13. Local results (2D) of the V-plus spur gear pair, rough (section in symmetry plane).

meshing. Incorporating the critical film thickness of $h_{cr}=1.70\text{ }\mu\text{m}$, the “rough” calculation always localized the tooth surface contact to the area of mixed friction. Therefore, because the developing solid body contacts absorbed part of the external load early, the minimum film thicknesses that appeared were also significantly larger than in the “smooth” calculation. However, the integral definition of the film thickness relative to the mean lines of the roughness profiles must be incorporated (see Fig. 5). The representation of the solid body contact load ratio as a function of the length of action reveals that approximately half of the external load from the solid body contacts was absorbed in the center. The characteristics of the maximum total pressures reflected the loading. The highest pressures were reached at the start of single meshing at point B, and as before, the decrease in the maximum pressures as of single meshing in point B was attributable to the better ratios of curvature in the contact. The maximum solid body contact pressures for the “rough” calculation displayed a uniform curve that only slightly dropped along the length of action.

Fig. 12 presents characteristics of the coefficient of friction and the friction force as a function of the length of action. The coefficient of friction is primarily influenced by the amount of

sliding in the motion between the tooth flanks, has a minimum in pitch point C because of the pure rolling motion there, whereas it attains higher values before and after the pitch point. The friction force that is actually acting at the contact tends to follow the characteristic of the load. The sign change that is noticeable at the pitch point results from the transition from a negative to a positive sliding motion. As expected, the friction values for the “rough” calculation are higher than those for the “smooth” calculation, and this change becomes clear when analyzing the mean friction power, which is approximately 88 W for the “rough” and only approximately 48 W for the “smooth” calculation.

Furthermore, Fig. 12 plots the curves of the maximum and mean temperatures at the tooth surface contact. The mean temperature was calculated for the entire area of the lubricating gap that was being analyzed. The temperatures rise rapidly at the beginning of meshing, and then, following the curve of the sliding velocity, the temperatures nearly return to their initial values, up to the pitch point, and then rise again afterward. The highest maximum temperature level is reached at the end of the length of action. In contrast, the mean temperature reaches its maximum

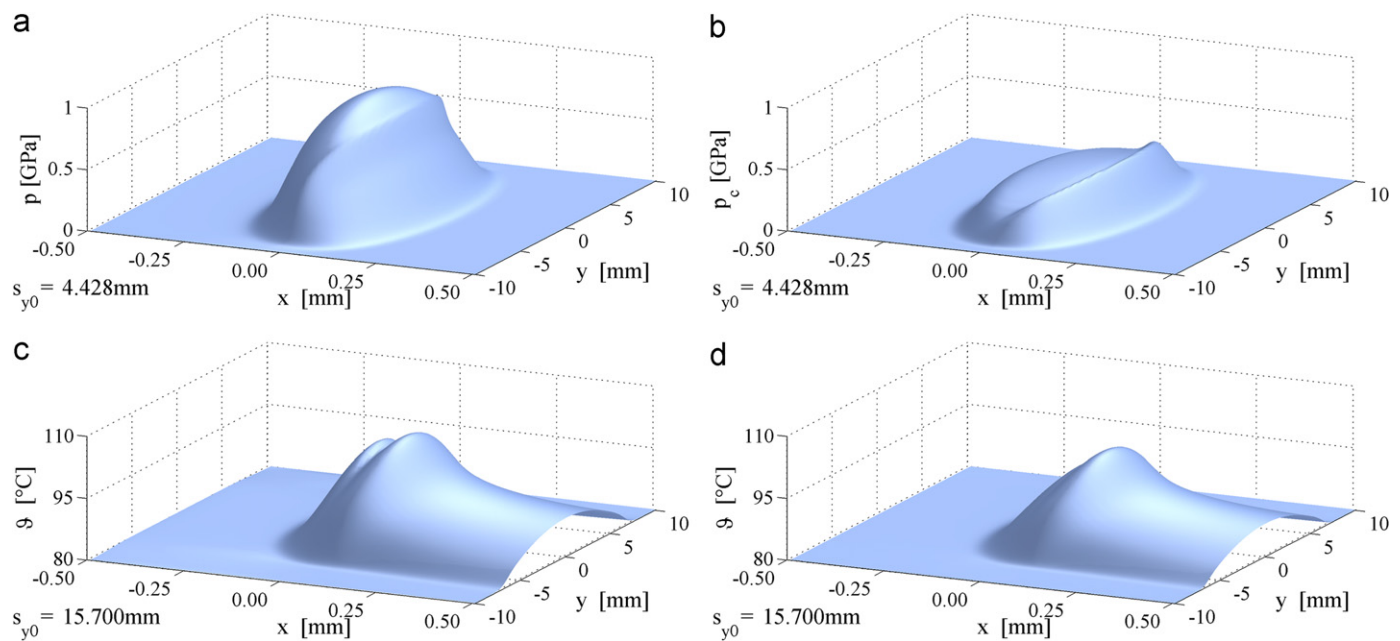


Fig. 14. Local results (3D) of the V-plus spur gear pair, rough, (a) pressure distribution (total) at point B, (b) pressure distribution (solid body contact) at point B, (c) temperature distribution at the mid lubricating gap at point E and (d) temperature distribution of the tooth flank at gear 1 point E.

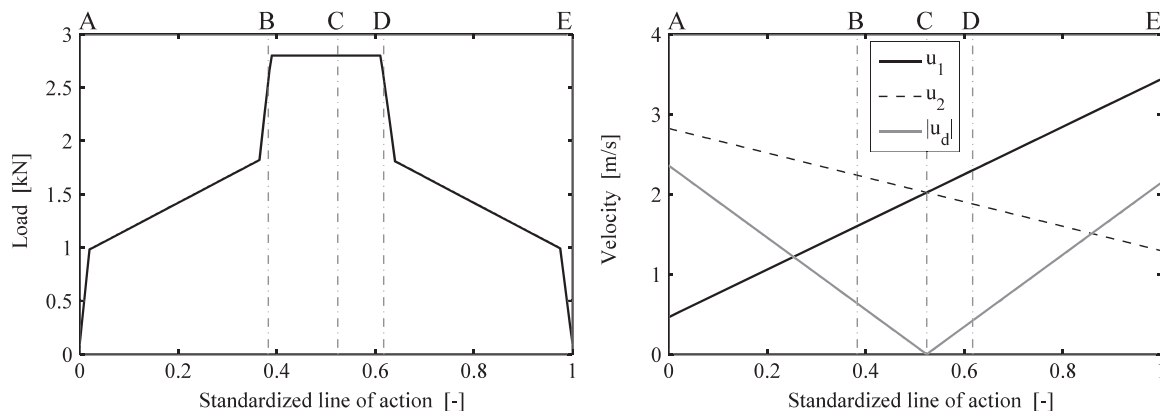


Fig. 15. Load and velocities of the spur gear pair with zero gearing ($x_{1,2}=0$).

value at the end of the single meshing at point D. Because the friction components are larger, a higher temperature level appears, when using the “rough” calculation.

Local results of the “rough” calculation for points A, B and E of the length of action are given in Figs. 13 and 14. The results that are presented for points A and B apply to the time immediately after the load has stopped increasing. The results presented for point E apply to the time that occurs immediately before the load begins to diminish (see Fig. 9). The 3D results in Fig. 14 are distorted in the x direction for better visualization. The characteristics of the pressures and film thicknesses display a form that is typical of elliptical elastohydrodynamic contacts. The size of the parallel gap that is produced by elastic deformations varies as a function of the load and the ratios of the curvature at the contact. The constrictions in the lubricating gap characteristics at the three outflow boundaries in the x and y directions are typical for this. The temperature characteristics that are presented clearly show that the maximum temperatures appear in the area of the

contact center, where the highest pressures and shear stresses are also reached. In the x direction of the contact, a temperature span develops along the lubricating gap. It is caused by the motion of the heated tooth flank areas out of the contact. The brief drop in temperature in the x direction is produced by constriction of the lubricating gap characteristic, which causes the compressed lubricant that is flowing into the cavitation region to expand. However, the largest heat sources that are caused by the boundary friction are not produced in the contact center but are produced the area of minimum film thickness.

A calculation of the analyzed spur gear pair with zero gearing ($x_1=x_2=0$, $a=a_d=112$ mm) and rough surfaces was performed as a first variation of the results presented thus far. The length of action increases to $g_x=19.14$ mm and produces a transverse contact ratio of $\varepsilon_x=1.621$, which is larger than that with V-plus gearing. All other parameters listed in Table 1 remain unchanged. The curves of the load and the tangential and sliding velocities are plotted in Fig. 15. Because the transverse contact ratio is larger,

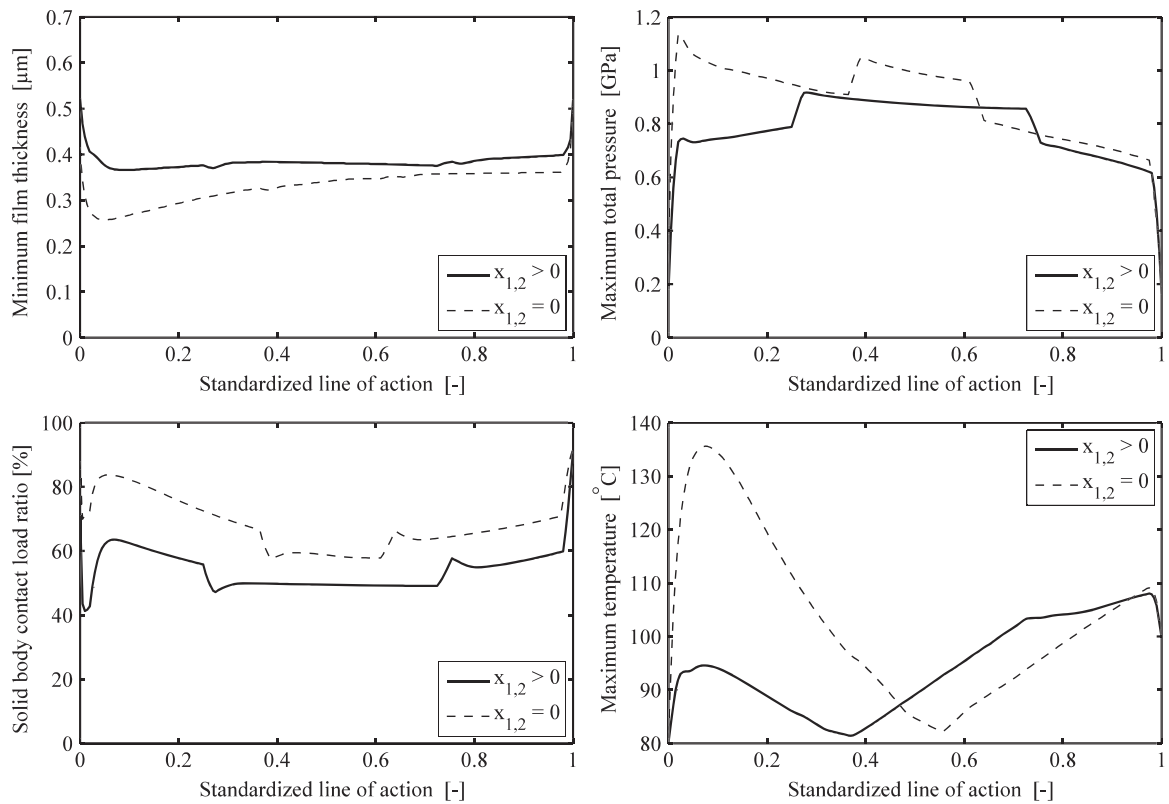


Fig. 16. Comparison of the results of the spur gear pair with V-plus and zero gearing.

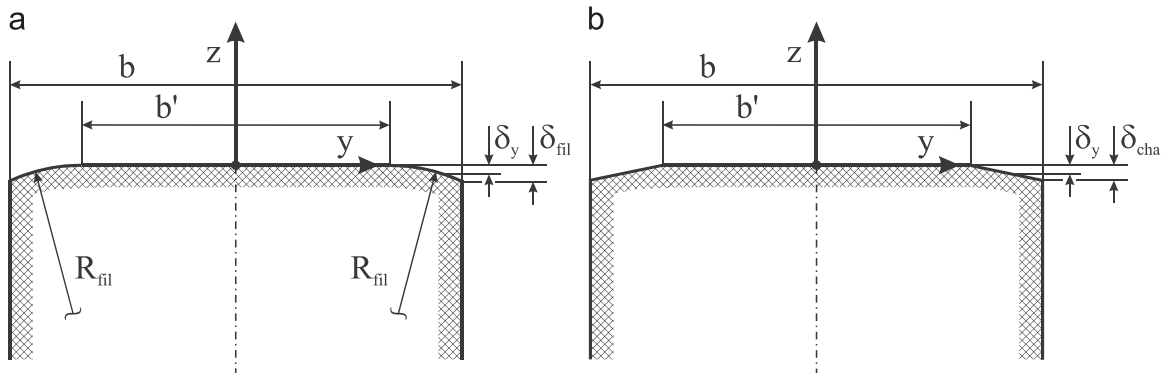


Fig. 17. Alternative tooth trace corrections in the direction of width, (a) fillet and (b) chamfer.

the relative share of single meshing is lower than that for V-plus gearing. However, the longer length of action with a higher sliding velocity causes meshing to commence at point A, and the gearing in this area is characterized by significantly poorer ratios of curvature at the tooth surface contact. Relative to the contact center, the reduced radius of curvature of the zero gearing at point A is 2.74 mm, but this value is 7.95 mm for the V-plus gearing.

Fig. 16 presents the most important results of zero gearing, when compared with V-plus gearing, as a function of the standardized length of action. The real lengths of action and the duration of the meshing of both gearings do not correspond. Because the ratios of curvature are poorer, the highest maximum pressures for the zero gearing are reached at the beginning of the meshing. While the peak values diminish along the length of action, these values only reach the level of the V-plus gearing after single meshing. The temperature level in the first region of double meshing is also significantly higher than that of the V-plus gearing. The higher sliding velocity between the tooth flanks is primarily responsible for this difference in temperature. The accompanying decrease in the hydrodynamic load capacity, which is due to temperature, leads to smaller minimum film thicknesses and larger solid body contact load ratios.

Calculations of the “rough” V-plus gearing with alternative tooth trace corrections in the direction of the width of the tooth flanks were performed at another stage of analysis. Specifically, the influence of fillets and chamfers on these calculations was studied (see Fig. 17). Unlike crowning, as is seen in Fig. 8, these tooth trace corrections alter the gap characteristic only at the edges of the teeth. Within a reduced width b' , the gap characteristic corresponds to a parallel gap. For the calculations, peak values at the edges of $\delta_{fil} = \delta_{cha} = 5 \mu\text{m}$ and a reduced width of $b' = 12 \text{ mm}$ were defined for both alternative corrections.

Fig. 18 presents the most important results for the different tooth trace corrections, which are plotted along the length of action. Only slight differences are detectable within the curves of the minimum film thickness and the solid body contact load ratio. The largest minimum film thicknesses are obtained by utilizing the calculation with the tooth trace correction that is designed as a fillet. However, this type of correction is also characterized by having the highest solid body contact load ratios. Greater differences arise between the curves of the maximum pressures and temperatures where the peak values from the calculation with filleting are significantly lower than those with crowning and chamfering.

Fig. 19 presents local results in the y direction of the contact for points B and E of the length of action, which explain these characteristics. The maximum pressures and temperatures and the smallest film thicknesses are produced at the crowning of the contact center. However, these extremes appear in the alternative tooth trace corrections as consequences of constrictions in the lubricating gap curve at the transitions between a parallel gap and the incipient tooth trace correction. The lowest maximum pressures and temperatures are detectable, as seen in Fig. 18, in the curves of tooth trace correction with filleting. The better performance is produced by the higher hydrodynamic load capacity of the parallel gap that is present in the region of the contact center. Furthermore, the transition between a parallel gap and the incipient tooth trace correction is consistently designed geometrically, and this property produces significantly better performance than chamfering. These statements apply only to parallel spur gears. Misalignments between the spur gears can cause severely one-sided loading of the contact, especially for the two alternative tooth trace corrections. In this application, crowning is better than filleting and chamfering.

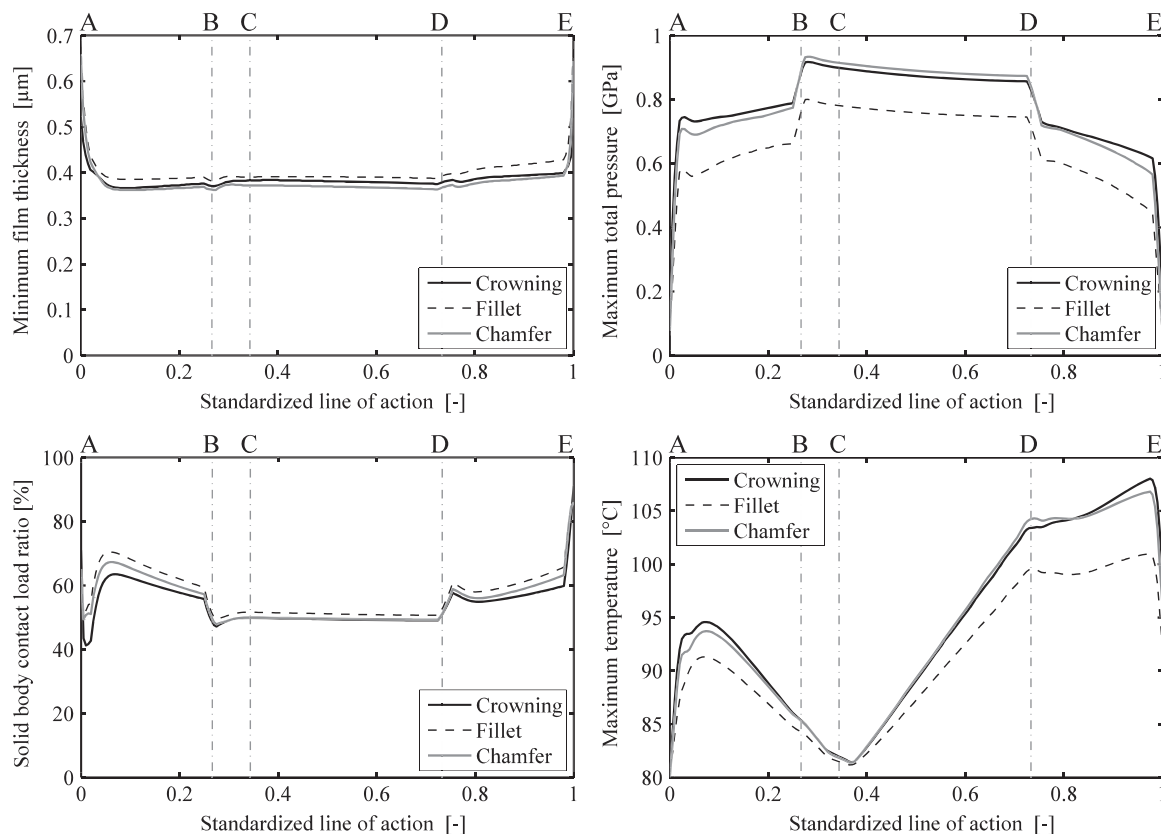


Fig. 18. Results of the V-plus spur gear pair, comparing different tooth trace corrections.

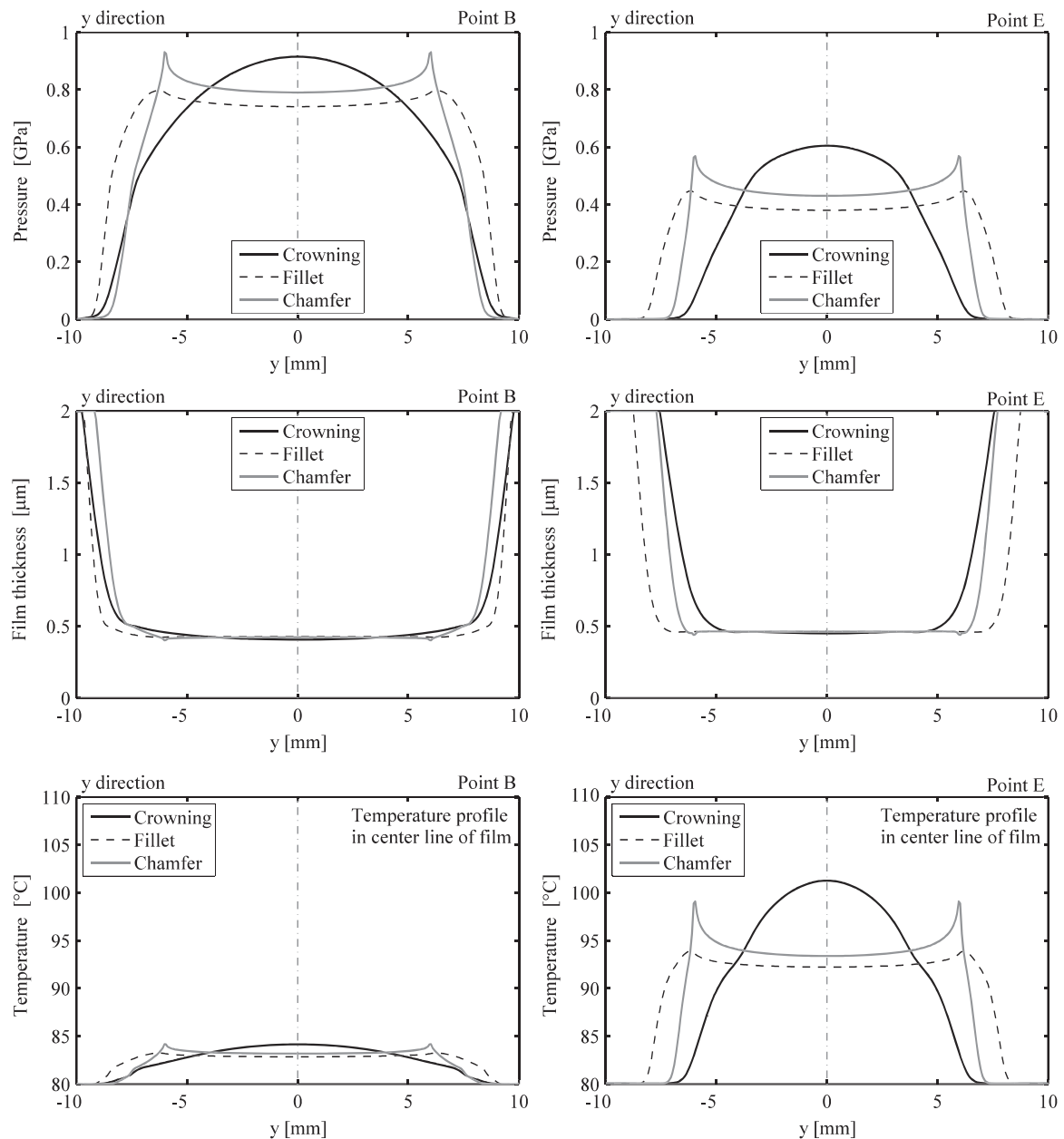


Fig. 19. Local results (2D) of the V-plus spur gear pair in the y direction (section in symmetry plane).

4. Conclusion

A model for calculating a transient, three-dimensional simulation of thermal elastohydrodynamic tooth surface contact of spur gears with involute gearing has been presented. In addition to allowing for mass conserving cavitation, non-Newtonian flow and the real involute characteristics of the tooth flanks, models that ascertain states of mixed friction and microhydrodynamic effects are integrated. A straight spur gear pair with positive addendum modification was selected for a demonstration calculation. First, the influence of the surface roughness asperities on the tribological properties at the tooth surface contact was analyzed. As expected, the friction and temperature curves were the greatest distinctions between the “rough” and “smooth” calculations. However, increases in the maximum pressures at the contact were not detected in the demonstrated calculation. Using another variation, the results of calculations

of the “rough” spur gear pair with and without an addendum modification have been discussed, which had a significant influence on the tribological processes at the tooth surface contact. Higher pressures and temperatures, smaller minimum film thicknesses and larger solid body contact load ratios appeared in gearing without addendum modification. The significantly poorer ratios of curvature and higher sliding velocities at the tooth surface contact at the beginning of meshing were identified to be crucial to these properties. Furthermore, the influence of different tooth trace corrections in the direction of width of the rough gearing was analyzed. As expected, the presented results clearly demonstrated that the form of the tooth trace correction has considerable influence on the load capacity of the tooth surface contact. A tooth trace correction that was designed as a fillet proved to be good, but only if significant misalignment of the gear wheels can be ruled out.

References

- [1] Dowson D, Higginson GRA. Numerical solution to the elasto-hydrodynamic problem. *Journal of Mechanical Engineering Science* 1959;1:6–15.
- [2] Wang KL, Cheng HSA. Numerical solution to the dynamic load, film thickness and surface temperatures in spur gears, part I analysis. *Journal of Mechanical Design* 1981;103:177–87.
- [3] Wang KL, Cheng HSA. Numerical solution to the dynamic load, film thickness and surface temperatures in spur gears, part II results. *Journal of Mechanical Design* 1981;103:188–94.
- [4] Vichard JP. Transient effects in the lubrication of hertzian contacts. *Journal of Mechanical Engineering Science* 1971;13:173–89.
- [5] Lin ZG, Medley JB. Transient elastohydrodynamic lubrication of involute spur gears. *Wear* 1984;95:143–63.
- [6] Oster P. Die Beanspruchung der Zahnflanken unter elastohydrodynamischen Bedingungen. *Tribologie und Schmierungstechnik* 1986;33:171–83.
- [7] Larsson R. Transient non-Newtonian elastohydrodynamic lubrication analysis of an involute spur gear. *Wear* 1997;207:67–73.
- [8] Wang Y, Li H, Tong J, Yang P. Transient thermoelastohydrodynamic lubrication analysis of an involute spur gear. *Tribology International* 2004;37:773–82.
- [9] Li S, Kahraman AA. Transient mixed elastohydrodynamic lubrication model for spur gear pairs. *Journal of Tribology* 2010;132:011501.
- [10] Akbarzadeh S, Khonsari MM. Performance of spur gears considering surface roughness and shear thinning lubricant. *Journal of Tribology* 2008;130:021503.
- [11] Castro J, Seabra J. Global and local analysis of gear scuffing tests using a mixed film lubrication model. *Tribology International* 2008;41:244–55.
- [12] Holmes MJA, Evans HP, Snidle RW. Analysis of mixed lubrication effects in simulated gear tooth contacts. *Journal of Tribology* 2005;127:61–9.
- [13] Ren N, Zhu D, Chen WW, Liu Y, Wang QJA. Three-dimensional deterministic model for rough surface line-contact EHL problems. *Journal of Tribology* 2009;131:011501.
- [14] Zhu D, Ren N, Qiang QJ. Pitting life prediction based on a 3D line contact mixed EHL analysis and subsurface von Mises stress calculation. *Journal of Tribology* 2009;131:041501.
- [15] Hamrock BJ, Dowson D. Isothermal elastohydrodynamic lubrication of point contacts. Theoretical Formulation. *Journal of Lubrication Technology* 1976;98:223–9.
- [16] Yang P, Wen SA. Generalized Reynolds equation for non-Newtonian thermal elastohydro-dynamic lubrication. *Journal of Tribology* 1990;112:631–6.
- [17] Venner CH. Multilevel solution of the EHL line and point contact problems. PhD thesis. University of Twente; 1991.
- [18] Wijnant YH. Contact dynamics in the field of elastohydrodynamic lubrication. PhD thesis. University of Twente; 1998.
- [19] Chevalier F, Lubrecht AA, Cann PME, Colin F, Dalmaz G. Film thickness in starved EHL point contacts. *Journal of Tribology* 1998;120:126–33.
- [20] Solov'yev S. Reibungs- und Temperaturberechnung an Festkoerper- und Mischreibungskontakten. PhD thesis. Otto von Guericke University Magdeburg; 2006.
- [21] Scholz U. Instationaere Berechnung geschmierter Reibkontakte. PhD thesis. Otto von Guericke University Magdeburg; 2008.
- [22] Elrod HGA. Cavitation algorithm. *Journal of Lubrication Technology* 1981;103:350–4.
- [23] Mostofi A, Gohar R. Elastohydrodynamic lubrication of finite line contacts. *Journal of Lubrication Technology* 1983;105:598–604.
- [24] Park TJ, Kim KW. Elastohydrodynamic lubrication of a finite line contact. *Wear* 1998;223:102–9.
- [25] Liu X, Yang P. Analysis of the thermal elastohydrodynamic lubrication of a finite line contact. *Tribology International* 2002;35:137–44.
- [26] Sun H, Xiaoyang C. Thermal EHL analysis of cylindrical roller under heavy load. In: *Proceedings of the IUTAM symposium on elastohydrodynamics and microelastohydrodynamics in Cardiff*; 2006. p. 107–20.
- [27] Bobach L, Bartel D, Deters L. Simulation elastohydrodynamischer Kontakte in Rollenlagern. VDI-Reports no. 2069. Duesseldorf: VDI Verlag; 2009. p. 53–65.
- [28] Townsend DP. *Dudley's Gear Handbook*. 2nd ed. New York: McGraw-Hill; 1992.
- [29] Bartel D. *Simulation von Tribosystemen—Grundlagen und Anwendungen*. 1st ed. Wiesbaden: Vieweg+Teubner Verlag; 2010.
- [30] Eyring H. Viscosity, plasticity and diffusion as examples of absolute reaction rates. *Journal of Chemical Physics* 1936;4:283–91.
- [31] Liu X, Jiang M, Yang P, Kaneta M. Non-Newtonian thermal analyses of point EHL contacts using the Eyring model. *Journal of Tribology* 2005;127:70–81.
- [32] Dowson D, Higginson GR. *Elasto-Hydrodynamic Lubrication*. 2nd ed. Oxford: Pergamon Press; 1977.
- [33] Rodermund H. Extrapolierende Berechnung des Viskositätsverlaufs unter hohen Druecken. *Schmieretechnik und Tribologie* 1980;27:3–5.
- [34] Patir N, Cheng HS. Application of average flow model to lubrication between rough sliding surfaces. *Journal of Lubrication Technology* 1979;101:220–30.
- [35] Larsson R, Andersson O. Lubricant thermal conductivity and heat capacity under high pressure. In: *Proceedings of the Institution of Mechanical Engineers, IMechE*, vol. 214; 2000. p. 337–47.
- [36] Bobach L. Simulation dynamisch belasteter Radialgleitlager unter Mischreibungsbedingungen. PhD thesis. Otto von Guericke University Magdeburg; 2008.
- [37] Mihailidis A, Retzepis J, Salpistis C, Panajiotidis K. Calculation of friction coefficient and temperature field of line contacts lubricated with non-Newtonian fluid. *Wear* 1999;232:213–20.